DRAFT RESEARCH REPORT 003 · FL-TR-2026-03

Show Your Work

Proof-carrying commits made an unattended coding factory's second night more useful. Task cards made building cheaper, not the night.

AUTHORS	DATE	WRITER · REVIEWER	STATUS
Infected Research / FrackLabs	2026-11-10	Grok 4.7 · Claude Opus 5.5	Draft · merged and deployed

Draft. All changes are merged onto master and deployed. Value labels and engineer-hours are AI estimates; value counts once the changes stay live.

ABSTRACT

We ran fourteen unattended coding-agent lanes on a private production codebase for one night, with the harness changed in response to round 1: every commit had to carry a proof test, prompts asked agents to reproduce before fixing, and a reviewer judged each commit as it landed. Against round 1, merged real fixes rose from 77 to 105, minor commits fell from 18% to 8%, and token cost per real fix held at about $5. Within the night we compared two arms of seven lanes. Free lanes hunted for their own work in one area each; scout-directed lanes built task cards that a read-only scout proposed. Cards made building 2.9× cheaper, but scouting, idle time and conflicts between card lanes cancelled the saving: on the pre-registered measure, cost per non-minor commit merged onto master, the arms came out at $4.18 and $4.27. Operator approval of cards rejected nothing. What improved was what each commit had to prove, not who chose the work.

HIGHLIGHTS

1. **More real fixes.** 105 merged commits fix a bug a real input can trigger, up from 77 in round 1 (+36%).
2. **Half the noise.** Minor commits fell from 18% to 8% of what was merged. Round 1 had 14 “fixes” for bugs no input could reach; this time the classifier found 0 and a blind spot-check one overstated claim in 16.
3. **Proof before review.** Every commit had to name a test that fails before the change and passes after. 144 of 147 passed the gate and 95% of those survived review (round 1: 89%).
4. **Same price per fix.** About $5.22 of model tokens per real fix, against $5.32 in round 1. The night bought more fixes, not cheaper ones.
5. **Cards: cheaper to build, not to run.** Scout-written task cards made building 2.9× cheaper per commit, but scouting cost more than building. All-in, the two arms finished 2% apart.
6. **Approval was a rubber stamp.** The operator accepted every card he reviewed and declined none. The human judgement that changed outcomes was at merge, not at approval.

1 Against round 1

ROUND 1 AND ROUND 2 ON THE SAME DEFINITIONS

	Round 1	Round 2	
Commits merged onto master	110	131	both runs one night, fourteen lanes
Real bugs fixed	77	105	a real input triggers the old behaviour
Minor share	18%	8%	worked, but nobody would have missed it
Fixes for unreachable bugs	14 of 100	0 of 105	classifier; spot-check found 1 overstated in 16
Survived review	89%	95%	of commits that passed the gate
Tokens, list price	$410	$548	writing, review, scouting
Per real fix	$5.32	$5.22	flat

The labels for both rounds come from the same model and classes, but from separate classification runs with slightly different prompts, so small differences should not be read closely. The large ones, the minor share and the unreachable fixes, also show in the review and spot-check results. Round 2 cost more in total because it ran more passes on a pay-per-token account after the subscription quota ran out; per real fix the price did not move. We cannot say which of the harness changes did the work: they shipped together.

2 Summary of findings

1. **The round got better, in both arms.** Minor commits were 6% of free and 10% of card commits, against 18% in round 1. The changes behind it applied to both arms: the proof requirement, reproduce-first prompting, the reviewer's value classes in the prompt, review after every commit, and test checkouts that no longer failed for lack of a config file (80 such failures down to about 21).
2. **Specifying the work made building cheap.** A card pass took a median 7 minutes and re-read 1.4M cached tokens; a free pass took 14 minutes and re-read 5.0M.
3. **The saving moved into scouting.** Scouting the card areas cost $148, more than the $91 spent building cards. $62 of it produced no cards: hung sessions, a run started in the wrong directory, and a run cut off by the quota.
4. **No difference all-in.** Per non-minor commit merged onto master, writing plus scouting cost $4.18 for free lanes and $4.27 for card lanes, +2%, far below the 30% we set in advance.

5. **A shared queue caused merge conflicts.** All 20 commits that did not apply automatically came from card lanes, which drew from one queue over seven areas. Free lanes, one area each, had none.
6. **Approval did not select.** The operator accepted 76 cards and declined none; 18 were never reviewed. The arm was scout-directed in practice. Waiting for approval left card lanes idle: they ran 12.4 lane-hours against 20.2.
7. **Cards shaped the kind of work, and sometimes the proof.** The scout was told to look for security gaps and missing behaviour tests, and card lanes produced 24 security fixes and 17 tests (free: 13 and 1). In the blind spot-check, 8 of 8 free proofs and 4 of 8 card proofs were genuine; some cards asked for a proof the change could not honestly give.

3 Method

The run used the factory from the first technical report (FL-TR-2026-01) with the changes listed below. Fourteen lanes ran in parallel, each in its own git worktree and branch from the same master commit (`88bf479c`). Each pass made at most one small commit. A gate checked every commit: Python compiles, JavaScript parses, forbidden files untouched, size limits, related tests and every Node test against the base, and a `Proof:` line in the message. Claude Opus 5.5 reviewed every commit that passed the gate before the lane continued: *keep*, *fix* (kept, and the next pass must fix what the reviewer named), or *drop* (removed from the branch). The writer was Grok 4.7 throughout.

Arms. Round 1's fourteen focus areas were split at random (seed 20261109) into two sets of seven. Free lanes got one area each, with round 1's focus text and a note that the obvious bugs had been fixed. Scout-directed lanes shared one queue of cards from the other seven areas. Before and during the run, a scout (Grok 4.7, read-only) read each card area and proposed cards: a title, a class, the files it had read, a brief, and the proof that would show the card done. The design called for the operator to accept or decline each card; in practice he accepted every card he reviewed. A card lane took the oldest accepted card, worked it until a commit survived the gate and review, then took the next. Two failed passes on one card marked it failed.

Changes since round 1, applied to both arms: the `Proof:` requirement; prompt changes (reproduce before fixing, the reviewer's value classes, a turn budget, no `git stash`); uncommitted leftovers saved as patches instead of stashed (round 1's shared stash swapped work between lanes); each pass shown its own branch history; a sanitized copy of the configuration file (secrets replaced) in every checkout, which cut tests that fail only in a worktree from 80 to about 21; a 25-minute pass limit and a watchdog for passes that stop producing output; and a detector that stops every lane when most of them fail together.

Pre-registration. The question, outcomes and the threshold for calling a difference were written before the run (`research/round2/criteria.md`). The primary outcome is writing and scouting cost at list price per non-minor commit integrated onto master; a difference of 30% or more, with at least 40 gate-passed commits per arm, counts as a result.

Integration. After the run every surviving commit was cherry-picked onto master in commit order. 117 applied cleanly; the operator's session resolved the 20 conflicts by hand. Six were not merged: three whose fix master already had from another lane (their tests pass on the merged code), two duplicate tests, and one folded into another commit. One was merged in part. The full suite showed no new failures against master, and the result was deployed and restarted the same day. Eight of the merged commits added known-failing tests that pinned open safety-filter gaps; the operator's session closed those gaps the same day, and the tests now pass.

Value labels and spot-check. After the run Claude Opus 5.5 classified all 137 surviving commits in random order, with lane, arm and card removed, from the message, the diff and the reviewer's reason. Classes are round 1's: security or safety, user-facing bug, reliability, behaviour test, minor. Four independent checkers, blind to arm and verdicts, took 16 commits (8 per arm, seed 20261110), ran each commit's proof test on the parent with only the new tests applied and again on the commit, and gave their own class. They agreed with the classifier on 14/16 classes and 15/16 real-bug calls.

Costs are token counts from the harness's session store priced at public list rates (Grok 4.7: $2 input, $0.50 cached input, $6 output and reasoning per million tokens; Claude Opus 5.5: $4, $0.20, $20, cache writes $5). The subscription charged nothing per token; the API account did. List prices make the arms and rounds comparable.

4 How the night went

Lanes started at 00:14 UTC on 10 November. The card lanes stopped at once on the first start: their queue lookup read a per-lane directory instead of the shared one. They were fixed and restarted at 00:18. The scout ran ahead of the builders, and the operator accepted cards in batches. From 02:19 card lanes began stopping with empty queues. At 02:45 the Grok subscription quota ran out; most lanes failed within eight minutes and the detector stopped all fourteen. At 06:13 the run resumed on the pay-per-token xAI account with the same model: free lanes immediately, card lanes after a new scout run and acceptance, from 06:34. At 07:15 the operator stopped the factory; lanes finished their passes by 07:35. Free lanes ran 20.2 lane-hours of passes, card lanes 12.4.

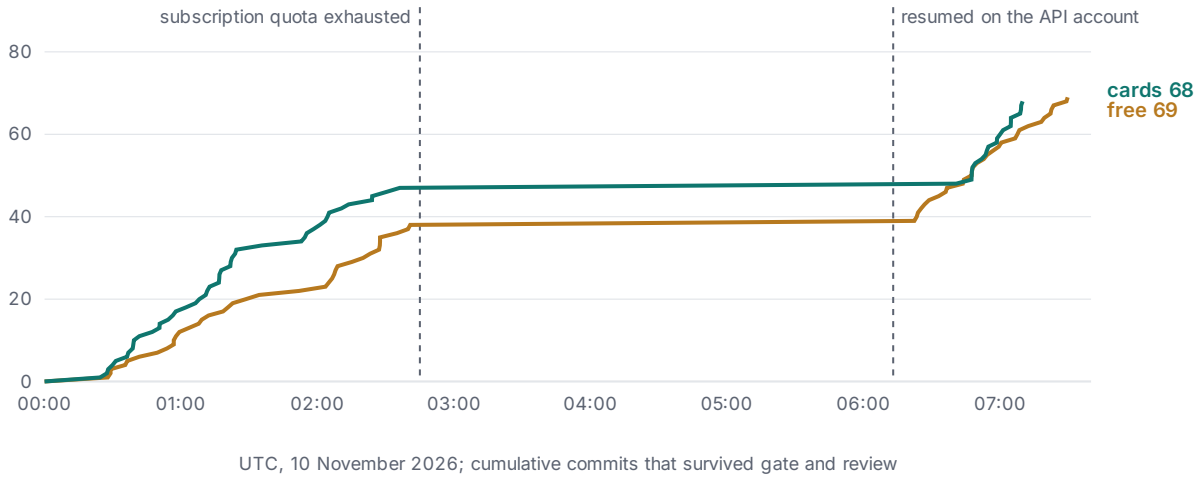


Figure 1. Commits that survived the gate and review, by arm, over time. Card lanes flatten when the queue empties; both stop when the quota runs out.

5 Two arms: free and scout-directed

RESULTS PER ARM

	Free	Cards	
Passes	**87**	**85**	including 13 and 12 with no commit
Passed the gate	**71**	**73**	3 and 0 failed it
Survived review	**69**	**68**	9 and 13 with a fix request; 2 and 5 dropped
Applied without conflict	**69**	**48**	conflicts: 0 and 20, resolved by hand
Merged onto master	**69**	**62**	non-minor: 65 and 56
Writing	**$271**	**$91**	list price
Scouting	**—**	**$148**	$62 produced no cards
Review	**$12**	**$13**	Opus, every commit
Primary: cost per non-minor commit merged	**$4.18**	**$4.27**	writing + scouting; cards +2%

The difference is far below the pre-registered 30%. Before the conflicts were resolved by hand, the stand-in measure (only commits that applied automatically) gave $5.43 against $4.18; hand resolution recovered most of the card arm's conflicting work. Leaving out the $16 spent by the scout run started in the wrong directory, a setup bug, gives the card arm $3.97.

Card lanes were idle for long stretches waiting for approved cards. A scout that keeps ahead of the builders, without a human approval step, would spend more on building and scouting but could produce more in the same night.

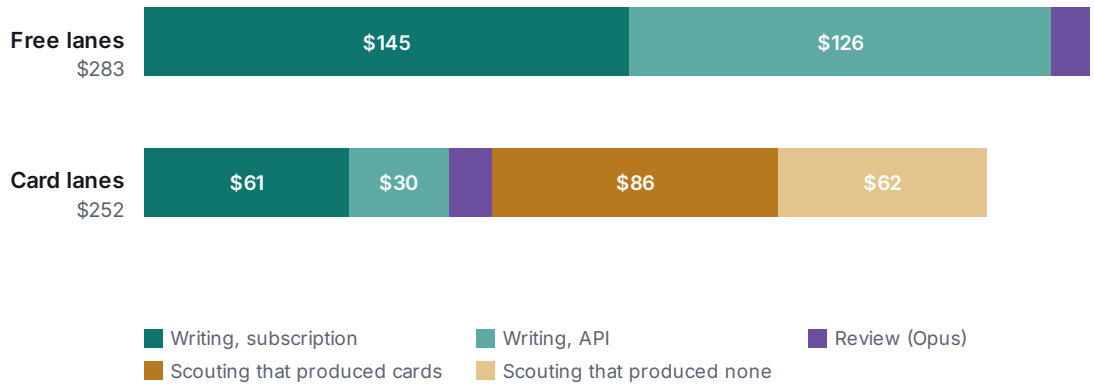


Figure 2. Spend per arm at list prices. Card building is cheap; scouting is most of the card arm's cost.



Figure 3. The card pipeline. Approval passed every reviewed card; conflicts between card lanes were resolved by hand at merge.

6 What the commits were worth

Blind classification found 64 of 69 free commits and 44 of 68 card commits fix behaviour a real input could trigger. The card arm's lower share comes from its 17 behaviour tests, which add tests for real decisions without a production change; the scout was told to propose them. The card arm found more security and safety problems (24 against 13), which the scout was also told to look for.

The classifier's rough estimate of engineering time to find, fix and test each non-minor change totals 187 hours for free lanes and 156 for card lanes: $1.51 and $1.62 of total spend per estimated engineer-hour. These are model judgements, not measured effort.

The spot-check found all 8 sampled free commits fixed real bugs, with proofs that failed before and passed after and no overclaiming. Of 8 card commits, 6 fixed real bugs, 4 had proofs that tested behaviour, and 2 overclaimed. In one the proof line said a new test failed before the change,

but the commit only added a test, as the card asked. Two other card proofs failed before only because a new module or constant did not exist yet.

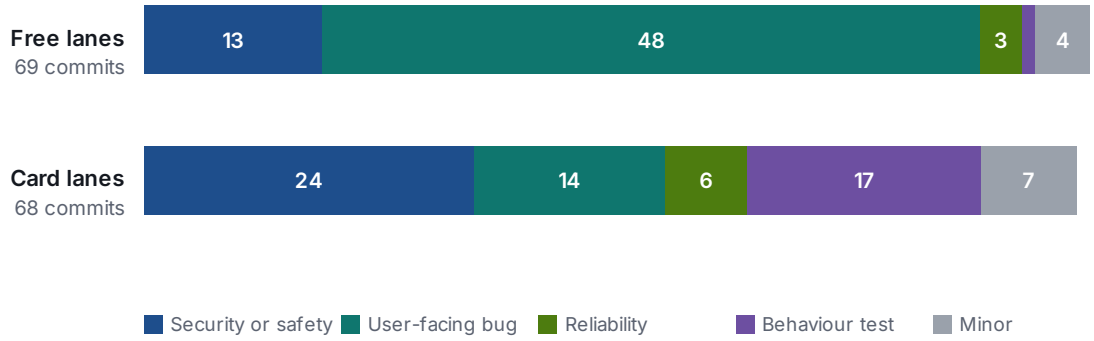


Figure 4. Surviving commits by value class, labelled blind to arm.

7 What we would change

- **Keep the proof requirement and check it mechanically.** The checkers' procedure, run the named test on the parent with only the commit's tests and again on the commit, is cheap enough for the gate. It would have caught the false proof and the proofs that fail only because a symbol is new.
- **Drop card approval; review at merge.** Approval rejected nothing and idled lanes. The merge review caught duplicates, superseded fixes and a false proof.
- **Give lanes their own files, not just their own queue.** Partitioning by area kept free lanes conflict-free. Card lanes need the same: assign each card to the lane that owns its files, or hold a card while another lane has its files open.
- **Keep the scout ahead and make it cheaper.** Scouting cost more than building and a third of it produced nothing. A scout run needs its own stall watchdog and should stop proposing for areas that come back empty.
- **Do not let cards dictate the proof's form.** A card for a behaviour test should ask for a test of a named production decision, not a test that fails before.
- **Make agents fix what they find, not only pin it.** In both rounds the factory found safety-filter gaps and recorded them as known-failing tests instead of fixing them: seven in round 1, eight here. A human session fixed both sets afterwards. A card or review rule that sends a pinned gap back for a fix would close that loop.
- **Plan for the quota.** Throughput was set by the provider's quota. A run plan should state which account is next and at what cost before it starts.

8 Limits

- One night, one codebase, one writer model, seven lanes per arm. Each area appeared in only one arm, so a rich or poor area moves its arm's numbers.
- The round-1 comparison is before and after, not controlled: the code had already been mined once, and all harness changes shipped together.
- Value classes, real-bug calls and engineer-hours are AI judgements. The spot-check covered 16 of 137 commits.
- Merged is not the same as proven in production. Value counts once the changes stay live and are not reverted.
- The arms did not run for the same time: card lanes waited for cards. Primary outcomes are per commit, not per hour.
- Scouting for free-arm areas ($38, unused) is excluded from both arms but included in the round's token total.
- The operator's time accepting cards was not recorded. Since approval declined nothing, it bought nothing measurable.
- The last 1.4 hours ran on a different billing account with the same model. Costs at list price are unaffected.

9 Data

- Sanitized per-commit records, per-arm numbers, the round comparison, token accounting and spot-check results: `https://fracktron.com/research/show-your-work/data.json`.
- Pre-registration: `criteria.md` in the same directory. Application source, diffs, commit bodies, cards and agent transcripts are not published.
- As in round 1, security and safety changes are listed by category only.

A Appendix: every surviving commit

LANE	COMMIT	SUBJECT	REVIEW	CLASS	MERGED
cards-1	ae308709	Security or safety change (category-only public record) (card 1)	keep	Security or safety	by hand
cards-1	65758a25	Security or safety change (category-only public record) (card 49)	keep	Security or safety	no: superseded
cards-1	df6dfbb4	db: let a renamed login save its own DM opt-out (legacy NULL row blocked the insert) (card 38)	keep	User-facing bug	yes
cards-1	01275c99	Security or safety change (category-only public record) (card 12)	keep	Behaviour test	yes
cards-1	b7b3c3e9	studio: page channel VOD lists past 200 (older replays were unreachable) (card 75)	keep	User-facing bug	yes
cards-1	8f019fbd	Security or safety change (category-only public record) (card 98)	keep	Security or safety	yes
cards-2	7c2e4257	Security or safety change (category-only public record) (card 2)	fix	Security or safety	yes
cards-2	1335dfb0	moderation: identify the broadcaster by twitch user id, not login (external relay names collide) (card 44)	fix	User-facing bug	yes
cards-2	7423438a	Security or safety change (category-only public record) (card 53)	fix	Security or safety	yes
cards-2	2e8ef7f6	moderation: time out viewers when the publisher twitch id is missing (use the streamer's id) (card 10)	keep	User-facing bug	yes
cards-2	a564d255	chat reader: hide own lines by Twitch user id (renames leave old logins visible) (card 65)	keep	User-facing bug	yes
cards-2	635d998a	offline-queue: dedupe UUID Discord messages with sha256 (hash() changes after restart) (card 69)	keep	User-facing bug	yes
cards-2	71b0a1ca	Security or safety change (category-only public record) (card 92)	keep	Security or safety	by hand
cards-3	bcd56ca8	Security or safety change (category-only public record) (card 3)	keep	Security or safety	yes
cards-3	30277515	diag alerts: page a crash burst from oldest unread id (newest page skipped the rest) (card 46)	keep	Reliability	yes
cards-3	4f7695d9	Security or safety change (category-only public record) (card 52)	keep	Security or safety	yes
cards-3	dadef15a	db: refuse a login tied to two user ids (ambiguous newest timestamp) (card 37)	keep	Minor	yes
cards-3	80b09668	Security or safety change (category-only public record) (card 8)	keep	Behaviour test	yes
cards-3	e4e781b8	Security or safety change (category-only public record) (card 56)	keep	Behaviour test	yes
cards-3	8c8b0bc3	Security or safety change (category-only public record) (card 61)	keep	Behaviour test	yes
cards-3	f9ad1fe1	live watch: do not commit seed after a locked notify read (channels skipped forever) (card 68)	keep	Reliability	yes
cards-3	bb50c8f7	daily boards: return real rank past 20th (player_best was 0) (card 73)	keep	Minor	yes
cards-3	83874f9a	Security or safety change (category-only public record) (card 76)	keep	Security or safety	yes
cards-3	93237cc5	chat reactions: count each person once (history and phone keys doubled) (card 93)	keep	User-facing bug	yes
cards-3	64f3cf57	overlay: rebind enrollment to the new Twitch user id (switch left the old channel) (card 85)	fix	User-facing bug	by hand

LANE	COMMIT	SUBJECT	REVIEW	CLASS	MERGED
cards-3	3b3df866	overlay: refuse a Twitch id only while its holder still has it linked (stale row blocked the new owner) (card 78)	keep	User-facing bug	by hand
cards-4	350c9954	Security or safety change (category-only public record) (card 4)	keep	Security or safety	yes
cards-4	b5ec5558	Security or safety change (category-only public record) (card 45)	fix	Security or safety	yes
cards-4	d6d24f1b	diag: cap intake rate-limit maps after idle eviction (unbounded growth) (card 51)	keep	Reliability	yes
cards-4	64ef6267	db: re-raise failed tweet inserts (do not skip the cursor) (card 39)	keep	Reliability	yes
cards-4	84ef46ce	Security or safety change (category-only public record) (card 7)	keep	Behaviour test	yes
cards-4	b4a95d46	Security or safety change (category-only public record) (card 11)	keep	Behaviour test	yes
cards-4	e1982256	imgur: xfail trumpet, Harrisburg, liberally, and regimen (unanchored prefixes) (card 58)	keep	Behaviour test	no: duplicate of 297ce44c
cards-4	26061a30	chat: keep same-second messages when a snapshot merges (ISO fraction sorts before Z) (card 94)	keep	User-facing bug	yes
cards-4	6d32d14a	db: resolve login to newest numeric chat speaker (ISO timestamp outranks unix time) (card 89)	keep	Minor	by hand
cards-5	a9ce2d61	db: ignore null stream names in the live roster (NULL channel_name crashed the read) (card 42)	keep	Minor	yes
cards-5	c13369f3	Security or safety change (category-only public record) (card 59)	keep	Behaviour test	yes
cards-5	d2a3dd47	Security or safety change (category-only public record) (card 60)	keep	Behaviour test	no: duplicate of af7d2b5b; spot-check found its Proof line false
cards-5	797da0d1	overlay: reserve daily quota in the consume write (parallel checks overshot the cap) (card 67)	fix	Minor	yes
cards-5	5720ad6f	Security or safety change (category-only public record) (card 71)	keep	Security or safety	no: superseded
cards-5	1cd0f80e	chat: claim duplicate sends before any await (in-flight retries posted twice) (card 95)	keep	User-facing bug	by hand
cards-5	e6e52eaf	live-watch: rekey rows when a numeric channel id renames (released login stayed watched) (card 86)	keep	User-facing bug	yes
cards-5	1bcd97c0	Security or safety change (category-only public record) (card 79)	keep	Behaviour test	by hand
cards-5	0455e7cf	Security or safety change (category-only public record) (card 84)	keep	Behaviour test	by hand
cards-6	bcb09a59	Security or safety change (category-only public record) (card 6)	keep	Security or safety	yes
cards-6	2b6ec718	newdev canary: confirm Xid alerts by fault class (rising count never paged) (card 47)	keep	Reliability	yes
cards-6	3004c158	Security or safety change (category-only public record) (card 50)	fix	Security or safety	by hand
cards-6	f569cb5f	diag: truncate oversized uploads instead of rejecting them (signed-in crash reports were dropped) (card 41)	keep	Reliability	by hand
cards-6	d9dcedb5	imgur: xfail unanchored blocked stems inside unrelated words (curated posts dropped) (card 9)	keep	Behaviour test	yes
cards-6	3b228e9a	Security or safety change (category-only public record) (card 55)	fix	Behaviour test	yes
cards-6	255be0cc	Security or safety change (category-only public record) (card 62)	keep	Security or safety	yes

LANE	COMMIT	SUBJECT	REVIEW	CLASS	MERGED
cards-6	0fedb3c7	db: index username history login lookups (LOWER() was a full scan) (card 70)	keep	Minor	by hand
cards-6	5d708699	Security or safety change (category-only public record) (card 96)	keep	Security or safety	yes
cards-6	1edb65e9	Security or safety change (category-only public record) (card 100)	keep	Security or safety	yes
cards-6	46bcaaa7	Security or safety change (category-only public record) (card 80)	keep	Behaviour test	yes
cards-6	e1c98622	Security or safety change (category-only public record) (card 83)	keep	Behaviour test	by hand
cards-7	f32a7c81	Security or safety change (category-only public record) (card 43)	fix	Security or safety	yes
cards-7	a27a6e3f	Security or safety change (category-only public record) (card 48)	fix	Security or safety	yes
cards-7	33b20a5d	Security or safety change (category-only public record) (card 54)	fix	Security or safety	yes
cards-7	e118bbe3	Security or safety change (category-only public record) (card 40)	keep	Security or safety	yes
cards-7	ef41d5b6	Security or safety change (category-only public record) (card 66)	fix	Security or safety	yes
cards-7	4b0b8f67	Security or safety change (category-only public record) (card 74)	keep	Security or safety	yes
cards-7	50b8e2c3	Security or safety change (category-only public record) (card 97)	keep	Security or safety	no: superseded
cards-7	d8b8b8fc	studio: set can_manage when the owner passes their slug (mobile always sends ?streamer=) (card 99)	keep	User-facing bug	yes
cards-7	59cbb507	memory: invalidate corrected claims by user id after a rename (stored login misses the row) (card 87)	keep	User-facing bug	yes
cards-7	65caf590	Security or safety change (category-only public record) (card 77)	fix	Behaviour test	by hand
cards-7	5a28fbb9	Security or safety change (category-only public record) (card 81)	keep	Minor	no: folded into 65caf590's resolution (phrases added to the existing parametrized xfail)
cards-7	b3dae868	Security or safety change (category-only public record) (card 82)	keep	Behaviour test	by hand
free-1	8e87648c	auth: keep a Twitch grant login does not cover (email scope was not a superset)	keep	User-facing bug	yes
free-1	3c2a0a63	auth: keep a provider grant login scopes do not cover (length and strict-superset checks still dropped them)	keep	User-facing bug	yes
free-1	a2f876f3	Security or safety change (category-only public record)	keep	Security or safety	yes
free-1	62d1babc	auth: keep a broader grant when linking or redeeming an invite (identity scopes replaced it)	fix	User-facing bug	yes
free-1	8894b551	auth: let a Discord re-link replace a dead or guilds-less token (bot scopes kept it)	keep	User-facing bug	yes
free-1	ef0697ee	auth: serve the Spotify account a re-auth just authorized (lookup kept the previous one)	keep	User-facing bug	yes
free-2	df203d77	vision: reject style-mix names that are not slugs (pattern was unanchored)	keep	Minor	yes
free-2	cc6eeefd	vision: honor mixer scales down to 0.2 (renderer clamped them to 0.4)	keep	User-facing bug	yes
free-2	87944c25	vision: stop !lora from calling a closed set live (photos were skipped)	keep	User-facing bug	yes

LANE	COMMIT	SUBJECT	REVIEW	CLASS	MERGED
free-2	62939906	vision: keep archived collections hidden when saved (Save sent closed)	keep	User-facing bug	yes
free-2	24c62024	vision: reject high quality with a known face (it rendered as an identity edit)	keep	User-facing bug	yes
free-2	1e2b0327	vision: keep the chosen style when comparing (Compare toggled it off)	keep	User-facing bug	yes
free-2	831ab646	vision: reject style mixes the edit lane would drop (they rendered as a face edit)	fix	User-facing bug	yes
free-2	1183cdd2	vision: drop a submitted mix when quality or the mixer changes (it blocked the next mode)	keep	User-facing bug	yes
free-2	7fbf6b9f	Security or safety change (category-only public record)	keep	Security or safety	yes
free-3	3b6ec7c8	studio chat: replace a handshake that never opens (chat stayed on connecting)	keep	Reliability	yes
free-3	3496995d	agent-room: refuse a blank prompt rewrite (it approved the original)	keep	User-facing bug	yes
free-3	37b0ed70	studio chat: drop a bad frame instead of disabling send (composer stayed dead)	keep	Minor	yes
free-3	644b7815	studio chat: back off when chat requires sign-in (anonymous tabs hammered reconnect)	keep	User-facing bug	yes
free-3	8a254b7e	agent-room: typed answer replaces a checked single choice (both were sent)	keep	User-facing bug	yes
free-3	3062f77b	agent-room: ignore a second click on a shell prompt (allow and deny both went out)	keep	User-facing bug	yes
free-3	95fbdad2	agent-room: ignore a second click on an agent question (reply and reject both went out)	keep	User-facing bug	yes
free-3	a2ce7395	agent-room: ignore a second click on a proposal (approve and reject both went out)	fix	User-facing bug	yes
free-3	9aa29cad	agent-room: grey out proposal cards in a real browser (tagName is BUTTON)	keep	User-facing bug	yes
free-3	fc9e1de5	agent-room: ignore a second click on Cancel (both cancels went out)	keep	User-facing bug	yes
free-3	aa8c006b	studio chat: clip !title before a split emoji (lone surrogate painted as tofu)	keep	User-facing bug	yes
free-4	1381e3da	audio: peak-normalize edits to -1 dBFS (stop clipping after upmix)	fix	User-facing bug	yes
free-4	c8a4590c	audio: peak-normalize large gains to -1 dBFS (volumedetect clamps at 0)	keep	User-facing bug	yes
free-4	82f44866	audio: fade looped scene beds once (stop seam dips)	keep	User-facing bug	yes
free-4	aaee9d49	audio: fade long scene beds at the canvas end (stop hard cuts)	keep	User-facing bug	yes
free-4	64919674	quote_game: skip same-name distractors (identical buttons scored as misses)	keep	User-facing bug	yes
free-4	570b4253	quote_game: ignore a late skip or start (stop abandoned rounds)	keep	User-facing bug	yes
free-4	b5bce59a	Security or safety change (category-only public record)	fix	Security or safety	yes
free-4	71ec14e3	soundboard: restore S3 full-object download assertion (privacy commit dropped it)	fix	Minor	yes
free-4	c32ded6f	soundboard: send Range on full-object download test (assertion could not fail)	keep	Behaviour test	yes
free-4	5f2dd17e	Security or safety change (category-only public record)	keep	Security or safety	yes
free-4	4da110c4	quote_game: ignore a late preview (stop abandoned rounds)	keep	User-facing bug	yes
free-5	806c193c	studio db: ignore stale publish edges (replay rewound the live session)	fix	User-facing bug	yes

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free-5	c0f7a4e0	studio db: accept same-session publish reconnects (mirror stayed offline)	keep	User-facing bug	yes
free-5	287a32e9	studio db: persist gif-only chat messages (history replay dropped them)	keep	User-facing bug	yes
free-5	d135427c	Security or safety change (category-only public record)	keep	Security or safety	yes
free-5	9e3fbb27	studio db: scope reply quotes to the channel (shared bot ids leaked text)	keep	User-facing bug	yes
free-5	0b3f810a	studio bridge: route Twitch chat by room id (reused login leaked into the wrong room)	keep	User-facing bug	yes
free-5	abbbef56	Security or safety change (category-only public record)	keep	Security or safety	yes
free-5	20154aa1	Security or safety change (category-only public record)	fix	Security or safety	yes
free-5	893e7cbb	Security or safety change (category-only public record)	keep	Security or safety	yes
free-6	ca943320	tracker: count rare tags by channel, not snapshot (hourly samples hid them)	keep	User-facing bug	yes
free-6	5b24d526	tracker: treat search wildcards as literal characters (underscores hid real logins)	fix	User-facing bug	yes
free-6	aa8f6c66	tracker: put NOCASE before ESCAPE so prefix search seeks (full scan)	keep	Reliability	yes
free-6	e405b825	tracker: pin category pages to the game id (reused names showed the other game)	keep	User-facing bug	yes
free-6	ec30db74	tracker: link reused category names to each game id (lists opened the other game)	keep	User-facing bug	yes
free-6	f31707d0	tracker: link reused logins to each numeric id (lists opened the other channel)	keep	User-facing bug	yes
free-6	37c3e40b	tracker: link reused Kick slugs to each user id (lists opened the other channel)	keep	User-facing bug	yes
free-6	a93078fa	Security or safety change (category-only public record)	keep	Security or safety	yes
free-6	5397ce44	tracker: keep another id when a reused login is released (reject deleted their join row)	keep	User-facing bug	yes
free-6	743f99c5	tracker: seed YouTube search from the current login owner (reused logins mixed the previous channel)	keep	User-facing bug	yes
free-6	d46b4903	tracker: seed YouTube search after dropping released logins (old owners filled the limit)	keep	User-facing bug	yes
free-6	8098c4ec	tracker: ignore stale live rows in channel search (dead channels ranked as live)	keep	User-facing bug	yes
free-7	4c6c0e33	ytdownloader: keep scanning Imgur routes after gallery (nested URLs parsed as the route token)	keep	User-facing bug	yes
free-7	8fdbaf36	youtube-vods: hide Remove once an import has evidence (button posted a refusal)	keep	User-facing bug	yes
free-7	cb1e6633	youtube-vods: classify a private Twitch download as a visibility block (403 looked like auth or a deleted VOD)	keep	User-facing bug	yes
free-7	b0951444	Security or safety change (category-only public record)	keep	Security or safety	yes
free-7	cc85d4c3	clipper: reclaim a running job after its worker dies (the active-VOD index left that VOD stuck)	keep	Reliability	yes
free-7	10a803af	youtube-vods: retry a manual part when a segment URL expires (403/404 looked like a private or deleted VOD)	keep	User-facing bug	yes
free-7	41070058	youtube-vods: keep a split in progress out of completed uploads (part_count 0 looked finished)	keep	User-facing bug	yes
free-7	21ba7fef	Security or safety change (category-only public record)	keep	Security or safety	yes
free-7	45a01672	Security or safety change (category-only public record)	keep	Security or safety	yes
free-7	4fd40656	Security or safety change (category-only public record)	keep	Security or safety	yes

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free-7	7dd36b61	ytdownloader: refuse gif-host redirects to internal hosts (tenor URL could 302 to metadata)	keep	Minor	yes